VR. A Formal System.

An Axiom-Free Reconstruction of Arithmetic from Operations Alone

Vitaly Reznik

2026 — Version 1.1.0

Abstract

We present a formal system VR generated by three primitive *operations* $\{\varnothing, \to, t\}$. VR posits *no axioms of its own*: the four named principles A1–A4, given as axioms in earlier versions, are here shown to be *definitions or theorems* over the freely generated term-algebra — A2 *defines* implication, A1 collects two of its values together with a closure theorem, A3 follows from the *definition* of membership, and A4 is the *recursor* of the inductive term structure. The von Neumann natural numbers are the unfolding of the succession operator t from $\varnothing$, the nullary base operation; membership $\in$ is a *defined* relation, not a primitive; equality is Leibnizian, and coincides with structural term-identity by a theorem. VR is arithmetically equivalent to first-order Peano arithmetic (PA), under the schematic reading of the Leibnizian quantifier, whence consistency relative to ZF follows as a metatheoretic corollary. "Axiom-free" is meant relative to the type-theoretic framework: VR adds no postulate beyond the free term-algebra (the nullary base and unary succession) and its recursor — this is exactly the axiom-free [] profile of the Lean 4 companion (Zenodo DOI 10.5281/zenodo.20324240), which formalises Part I.

Keywords: operational reconstruction of arithmetic, axiom-free system, foundations of mathematics, Leibnizian equality, von Neumann ordinals, Peano arithmetic, consistency.

Version note (1.1.0). This is a minor-version revision of v1.0.3 (a change of thesis, not a patch: the title moves from "axiomatization" to "axiom-free reconstruction"). It (i) carries the operational reading ("there are only operations; $\varnothing$ is a nullary operation, not a posited thing") into the *body* of Part I — membership $\in$ is *defined* by recursion on t rather than presupposed, the gloss $t(x) = x \cup \{x\}$ is marked an abbreviation, F and $\varnothing$ are parallel nullary bases in two registers rather than one entity, and the scope of the propositional basis and the metatheoretic status of the quantifiers are made explicit; and (ii) **reframes the system as axiom-free** — the four named principles A1–A4 are reclassified as definitions and theorems over the free term-algebra, not postulates, matching the axiom-free Lean 4 companion. No statement is altered; only the status of A1–A4 is reclassified. The definition of $\in$ and the acyclicity Lemma are made explicit, the relative-consistency statement is marked metatheoretic, and the equality in the definition of $\in$ is identified as structural term-identity (related to Leibnizian equality by `Eq_to_vrEq` and its immediate converse). Earlier editorial notes (v1.0.1–1.0.3) are retained in Part IV.

Contents

Part I. VR — the formal system

1. Primitives

VR is generated by three primitive *operations*:

- the **nullary** operation $\varnothing$ (the base — the act taking no operands; *not* a posited thing that "is empty");
- the **unary** operation t (succession);
- the **binary** operation $\to$ (implication, on the logical layer).

Every object of VR is a *term* over this signature: $\varnothing$, $t(\varnothing)$, $t(t(\varnothing)), \dots$. There are only operations and the terms they compose; no object is posited beyond what these operations generate. (In the Lean 4 companion this is literal: the objects form an inductive type `VRObj` with constructors `base` and `succ`.) This free term-algebra — together with its recursor (A4 below) — is the system's *only* foundational commitment; everything else, including the principles A1–A4, is definition or theorem (§2).

Membership is defined, not primitive. The relation $\in$ used throughout is *not* a fourth primitive; it is defined by structural recursion on the generated term:

$$z \in \varnothing \;:=\; \mathsf{F}, \qquad z \in t(y) \;:=\; (z \doteq y) \;\vee\; (z \in y). \tag{1}$$

Here $\doteq$ is **structural term-identity** ("is z the same generated term as y?"), which is prior to and independent of the Leibnizian equality of Def. 2: $\in$ is built on structural identity alone, so *no circularity* arises. Structural identity implies Leibnizian equality (Lean: `Eq_to_vrEq`); the converse is immediate (instantiate the discriminating property $p := (x \doteq \cdot)$), so the two coincide — though only the direction actually used, `Eq_to_vrEq`, is recorded in the companion. Thus membership is decided by descending the term's t-structure and introduces no new primitive content. The familiar gloss $t(x) = x \cup \{x\}$ (used in A3) is likewise an *abbreviation*: $\cup$ and $\{\cdot\}$ are not primitives, and the substantive content of that gloss ($x \in t(x)$ and $x \subseteq t(x)$) is a *theorem* from (1) (§5). This keeps the "three operations" claim exact: $\in$, $\cup$, $\{\cdot\}$ are derived. (Lean: `mem` is (1); `A3_mem_self`, `A3_subset_succ` are theorems.)

2. The four named principles A1–A4 (definitions and theorems, not postulates)

Earlier versions stated A1–A4 as axioms. Under the operational reading each is a *definition* or a *theorem* over the freely generated objects; VR posits no axioms of its own. We keep the names A1–A4 for reference and mark the true status of each.

A2 (Implication) — a *definition*. $\to$ is the function $\{\mathsf{F}, \top\} \times \{\mathsf{F}, \top\} \to \{\mathsf{F}, \top\}$ given by the classical truth table: $\mathsf{F} \to \mathsf{F} = \top$, $\mathsf{F} \to \top = \top$, $\top \to \mathsf{F} = \mathsf{F}$, $\top \to \top = \top$. This is the definition of the operation $\to$; it postulates nothing (Lean: `impl` is this definition, its four values `rfl`).

A1 (Generativity) — *facts of A2 plus a closure theorem.* The two base implications $\mathsf{F} \to \mathsf{F} = \top$ and $\mathsf{F} \to \top = \top$ are simply two entries of the table A2 (hence hold by definition), with $\top := \mathsf{F} \to \mathsf{F}$. The substantive claim of A1 is a *closure* statement, and it is a *theorem*: the smallest set containing F and closed under $\to$ is all of $\{\mathsf{F}, \top\}$ — indeed $\mathsf{F} \to \mathsf{F} = \top$ yields $\top$, and $(\mathsf{F} \to \mathsf{F}) \to \mathsf{F} = \top \to \mathsf{F} = \mathsf{F}$ recovers F, so iterating $\to$ from F reaches both values; whereas the smallest set containing $\top$ and closed under $\to$ is $\{\top\}$ alone. The point is closure under iteration, not a single application. (Lean: `A1_F_reaches_both`, `A1_T_reaches_only_T` are these theorems; `A1_1`, `A1_2` the `rfl` facts.)

Note on F and $\varnothing$ (two registers, not one object). F is the base *value* of the logical register $\{\mathsf{F}, \top\}$; $\varnothing$ is the base *object* of the operational register. They are *parallel* nullary bases — one act with no operands in each register — written in correspondence but *not* identified as the same entity. (The Lean companion keeps them as distinct types `VRBool` and `VRObj`; the correspondence is used by no theorem — cf. the level-distinction for $\leftrightarrow$, §17.)

A3 (Succession) — a *theorem.* t is the primitive succession operation, total on the generated objects. The content ascribed to A3 — written as the gloss $t(x) = x \cup \{x\}$ — is exactly: for every x, $x \in t(x)$ and $x \subseteq t(x)$. By the definition (1) of $\in$ both *hold as theorems* ($x \in t(x)$ since $x \doteq x$; $x \subseteq t(x)$ since $z \in x \Rightarrow z \in t(x)$). A3 postulates nothing beyond the totality of the constructor t, which is part of the term-algebra.

A4 (Induction) — the *recursor.* If P is a property of objects and (i) $P(O_0)$ and (ii) $\forall x : P(x) \to P(t(x))$, then $P(O_n)$ for all n; equivalently, the O_n exhaust the objects generated from $\varnothing$ via t. In the typed setting this is precisely the *recursor* (eliminator) of the inductive term-algebra — a derived principle, not a postulate (§13).

Axiom-free, precisely. The system's only foundational commitment is the free term-algebra: the objects are exactly the terms generated by the nullary base $\varnothing$ and the unary succession t, with the associated recursor. Given that framework, A1–A4 add no postulational content: A2 is a definition, A1 and A3 are theorems, A4 is the recursor. Hence VR is **axiom-free** — meaning it adds *no axioms beyond the type-theoretic framework* (which itself furnishes the inductive type and its recursor, and with them the potential infinity of $\mathbb{N}$). This is exactly the [] profile certified by the Lean companion: `#print axioms` reports no `propext`, `Classical.choice`, or `Quot.sound` for any object of Part I, and the inductive type and recursor are kernel primitives, not axioms. *What Peano postulates as axioms, VR derives.* The claim is not that VR assumes nothing — it assumes the term-algebra framework — but that it introduces no axioms of its own.

3. The logical basis and its scope

$\{\mathsf{F}, \to\}$ is functionally complete for classical two-valued *propositional* logic over $\{\mathsf{F}, \top\}$. Derived connectives:

$$\neg x := x \to \mathsf{F}, \quad x \vee y := (x \to y) \to y, \quad x \wedge y := \neg(\neg x \vee \neg y), \quad x \leftrightarrow y := (x \to y) \wedge (y \to x).$$

Scope. This completeness concerns the *propositional* connectives only. The *quantifiers* appearing in definitions and proofs — the $\forall p$ of Def. 2 and the $\exists$ of P2 (§9) — are not part of this basis; they belong to the ambient (first-order) metatheory and are read *schematically* (see §10 and the note to Def. 2). The propositional layer of VR is two-valued and complete; the arithmetic is developed within the ambient quantified logic over the generated objects.

4. Definitions

Def. 1. $\top := \mathsf{F} \to \mathsf{F}$.

Def. 2. **Leibnizian equality.** $x = y \ := \ \forall p : \ p(x) \leftrightarrow p(y)$, p ranging over properties of objects. *Reading.* $\forall p$ is taken *schematically* — a schema over the (arithmetically definable) properties, matching the schema interpretation used for the PA-encoding in §10 — not as a genuine second-order quantifier. (A full second-order reading would make VR's equality second-order and the equivalence of §11 one with *second-order* PA, which is categorical and strictly stronger; the equivalence we prove, §11, is with *first-order* PA under the schematic reading. The $\leftrightarrow$ here is metatheoretic, distinct from the propositional operator of §3 — see §17. This $=$ relates to the structural identity $\doteq$ of (1) by `Eq_to_vrEq` and its immediate converse. Lean: `vrEq` over `VRObj→Prop`.)

Def. 3. **Distinctness.** $x \neq y := \neg(x = y)$.

Def. 4. $O_0 := \varnothing$. Def. 5. $O_{n+1} := t(O_n)$. Def. 6. The natural number n is identified with the ordinal O_n.

5. Acyclicity of $\in$, and $t(x) \neq x$

Lemma 1 (Antisymmetry and irreflexivity of $\in$)**.** *For all objects x, y: if $x \in y$ then $y \notin x$; in particular $x \notin x$.*

Proof. By induction on y (using (1)). Base $y = \varnothing$: $x \in \varnothing$ is F, vacuous. Step $y = t(w)$: suppose $x \in t(w)$ (i.e. $x \doteq w$ or $x \in w$) and, for contradiction, $t(w) \in x$. The lowering fact $t(a) \in b \Rightarrow a \in b$ (immediate induction on b; when $a \doteq w$ it uses $a \in t(a)$, reflexivity in (1)) gives $w \in x$. If $x \doteq w$, then $w \in w$, contradicting the hypothesis at w; if $x \in w$, then with $w \in x$ the hypothesis at w forbids $w \in x$. Either way contradiction. Irreflexivity is $y := x$ against itself. (Lean: `mem_asymm`, `not_mem_self`, axiom-free.) □

Proposition 1. *$t(x) \neq x$ for every x.*

Proof. The property "contains x" is $\top$ at $t(x)$ (since $x \in t(x)$, A3) and F at x (since $x \notin x$, Lemma 1); it distinguishes them, so $t(x) \neq x$ by Def. 3. □

6. The von Neumann numbers

$$O_0 = \varnothing, \quad O_1 = \{\varnothing\}, \quad O_2 = \{\varnothing, \{\varnothing\}\}, \quad O_3 = \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}, \ \ldots$$

Each O_n contains all preceding $O_0, \ldots, O_{n-1}$ (from (1) by induction; Lean: `O_mem_lt`).

7. Arithmetic

Addition $a + O_0 := a$, $a + t(b) := t(a + b)$; multiplication $a \times O_0 := O_0$, $a \times t(b) := (a \times b) + a$; exponentiation $a^{O_0} := O_1$, $a^{t(b)} := a^b \times a$. **Theorems** (by induction, A4): T1 commutativity, T2 associativity, T3 distributivity, T4 $O_1 + O_1 = O_2$. (On T1's structural cost and on T1–T4 not being needed for §11, see §16.)

8. Structure of the system

Layer	Content
Foundational commitment	the free term-algebra on $\{\varnothing \text{ (nullary)}, t \text{ (unary)}\}$ + recursor
Primitive operations	$\varnothing$, $\to$, t; $\in$ defined by (1); $\cup, \{\cdot\}$ abbreviations
Named principles	A2 (def. of $\to$), A1 (facts + closure thm), A3 (thm), A4 (recursor) — *no postulates*
Logic	classical two-valued propositional; quantifiers metatheoretic (§3)
Equality	Leibnizian, schematic (Def. 2); $= \equiv \doteq$ by theorem
Objects & arithmetic	von Neumann O_n; $+, \times, ^\wedge$ (A4)

Part II. Equivalence of VR and Peano arithmetic

9. Direction VR $\to$ PA

The five Peano axioms (P1 $0 \in \mathbb{N}$; P2 $\forall n \, \exists S(n)$; P3 $S(n) \neq 0$; P4 $S(n) = S(m) \Rightarrow n = m$; P5 induction) hold in VR under $\mathbb{N} \mapsto \{O_n\}$, $0 \mapsto O_0$, $S \mapsto t$.

- **P1, P2.** $O_0 := \varnothing$ is a term; t is total on the generated objects. (§14: in the typed setting these are facts of the syntax, not theorems.)
- **P3.** "contains an element" is F at $\varnothing$, $\top$ at $t(O_n)$; by Def. 3 they differ.
- **P4** ($t(x) = t(y) \Rightarrow x = y$). From $t(x) = t(y)$ and (1) the members coincide, giving ($x = y \vee x \in y$) and ($y = x \vee y \in x$). By Lemma 1: the case $x \in y \wedge y \in x$ is excluded by antisymmetry; the mixed cases $x = y \wedge y \in x$ and $x \in y \wedge y = x$ substitute to $x \in x$, excluded by irreflexivity. Only $x = y$ survives. (Lean: `P4_succ_inj` via constructor injectivity.)
- **P5** is A4.

10. Direction PA $\to$ VR

Encode VR objects as naturals by arithmetization (Gödel numbering): $\ulcorner \varnothing \urcorner := 0$, $\ulcorner t(x) \urcorner$ a computable function of $\ulcorner x \urcorner$, so $\ulcorner O_n \urcorner = n$. The defined $\in$ and the gloss-operations are primitive recursive on codes (Mendelson; Boolos & Jeffrey). A1–A2 are derivable in PA's logic; A3 is a primitive recursive relation; A4 is PA induction. Leibnizian equality is read as the *schema* over arithmetical formulas — the schematic reading of Def. 2 — which makes the target *first-order* PA.

11. Equivalence theorem

Theorem 1. *Under the schematic reading of Def. 2, VR and first-order PA are arithmetically equivalent: the $\mathbb{N}$-theoretic content of one corresponds bijectively to that of the other.*

The Lean companion realises this as a concrete structural isomorphism $O : \mathbb{N} \to$ `VRObj` with inverse, preserving $0, \mathrm{succ}, +, \times, ^\wedge$ (`Theorem_11_VR_PA`); §15.

12. Consistency relative to ZF (metatheoretic)

Corollary 1. *VR is consistent relative to ZF.*

Proof. PA is consistent relative to ZF (the von Neumann $\mathbb{N} \subseteq$ ZF realises the PA axioms — classical). By Theorem 1, a contradiction in VR would yield one in PA, hence in ZF; if ZF is consistent, so is VR. □

Status. A *metatheoretic* corollary via the equivalence; *not* part of the Lean formalisation. The formalisation verifies Part I's theorems by the kernel and is axiom-free in Lean's sense; the relative-consistency statement is classical metatheory cited here. Chain: VR $\leftrightarrow$ PA $\hookrightarrow$ ZF.

Part III. Open questions

(1) **Number systems.** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ over VR. *Status:* VR-Numbers (DOI 10.5281/zenodo.20272743), as operational superstructures over VR's naturals without posited ordered pairs.

(2) **Type theory.** VR as an initial t-algebra; t = `succ`, $\varnothing$ =zero, A4 = the `Nat` eliminator; Leibnizian vs. path equality (HoTT).

(3) **Tarski / semantic closure.** Whether a level hierarchy via iteration of t evades undefinability (a direction, not a result).

(4) **Proof assistant.** *Status:* completed in Lean 4 (DOI 10.5281/zenodo.20324240); Part IV.

Part IV. Notes from the Lean 4 formalisation

Part I is fully formalised in Lean 4 (DOI 10.5281/zenodo.20324240): every theorem is verified by the kernel, no mathlib arithmetic is used, and every object is axiom-free in Lean's sense. The notes mark where the formal layer is more revealing; several are now folded into the body (v1.1.0) and kept here for the record.

13. A4 is the recursor, not an axiom. VR-objects form an inductive type (`base`, `succ`); A4 is its recursor — derivable, not postulated.

14. P1 and P2 are absorbed by typing. O_0 is a well-typed term; succ is total. Both are facts of the syntax. Only P3–P5 retain content as theorems.

15. The Equivalence Theorem (§11) has a constructively stronger form. The metatheoretic equivalence is realised as an explicit type isomorphism with inverse, preserving all operations.

16. T1–T4 are not needed for the equivalence. The isomorphism $O(m+n) = \mathtt{vadd}(Om)(On)$ and its analogues close by induction on the right argument, using only the recursive symmetry of `Nat`/`vadd` etc. (T1 is structurally costlier than T2–T3: addition recurses in its second argument, so $O_0 + b = b$ and $t(a) + b = t(a + b)$ are separate lemmas — a cost the symmetric notation hides.)

17. $\leftrightarrow$ carries two meanings. In §3 the truth-functional biconditional on $\{\mathsf{F}, \top\}$ (`viff`); in Def. 2 the metatheoretic equivalence of assertions (`Iff` on `Prop`).

18. F and $\varnothing$ are two registers. The A1 correspondence between the logical base F and the operational base $\varnothing$ is, in the formalisation, a correspondence between distinct types (`VRBool`, `VRObj`); invoked by no theorem.

19. $\in$ is defined; equality is structural; acyclicity is structural. $\in$ is the recursive definition (1) (`mem`); the $=$ in it is the kernel's structural `Eq`; structural identity implies Leibnizian equality (`Eq_to_vrEq`), the converse being immediate, so they coincide (only `Eq_to_vrEq` is recorded, the converse not being needed); the set-content of A3 is a theorem (`A3_mem_self`, `A3_subset_succ`); acyclicity (Lemma 1) is by structural induction with no measure (`mem_asymm`, `not_mem_self`).

20. The system is axiom-free. Collecting §13 (A4 = recursor) with the reclassification of A2 (definition of $\to$), A1 (`rfl` facts + closure theorem) and A3 (theorem from (1)): VR has *no proper axioms*. `#print axioms` returns [] for every object of Part I (no `propext`, `Classical.choice`, or `Quot.sound`). The sole foundational commitment is the inductive term-algebra and its recursor — kernel primitives, not VR axioms. This is the precise sense of "axiom-free": VR adds nothing to the type-theoretic framework; what PA postulates, VR derives.

Acknowledgement of AI assistance

This preprint and its companion Lean 4 formalisation were prepared with the assistance of Claude (Anthropic), in the Variant A architecture used across the VR cycle: a human curator directing a model serving as architect and implementer. All mathematical content, definitions, theorems, and positions are due to the author, Vitaly Reznik. Version 1.1.0 (operational reading carried into Part I; the axiom-free reframing) was prepared with Claude Opus 4.8.

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